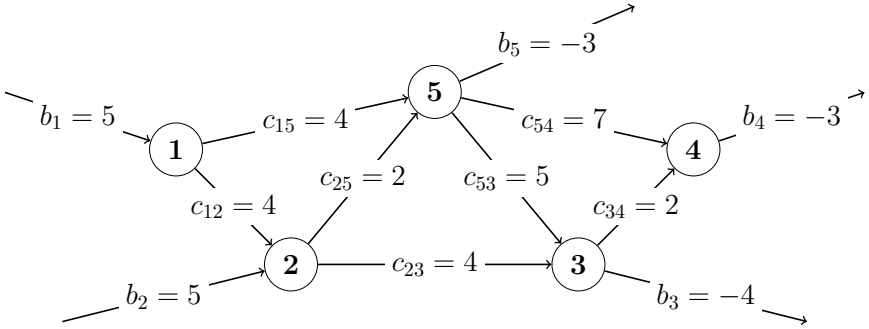
Exercise session 2

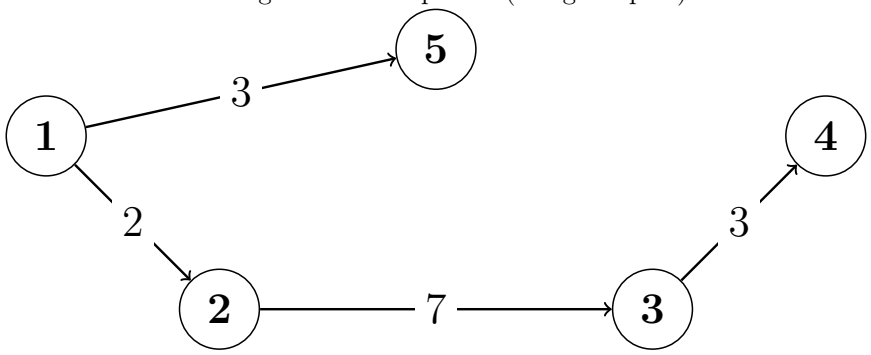
Problem 1

Find the minimum-cost flow in the network/graph consisting of nodes $\mathcal{N} = \{1, 2, 3, 4, 5\}$ and edges (bågar) $\mathcal{B} = \{(1, 2), (1, 5), (2, 3), (2, 5), (3, 4), (5, 3), (5, 4)\}$. The graph, with sources (källnoder), sinks (sänknoder), and edge cost marked are:



a) Given the order in $\mathcal{B}$, write down the incidence matrix (incidensmatrisen), $\tilde{A} \in \mathbb{R}^{5\times 7}$. Pose the problem of finding the minimum-cost flow as an LP on standard-form.

b) Show that the following solution is optimal (using Simplex).



c) Let $c_{53} = 3$. Show that the solution in b) is no longer optimal. Find an optimal solution (using Simplex). Start from the solution in b).

Problem 2

Given the matrix

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 3 & 1 \\ 3 & 3 & 4 & 2 \end{pmatrix}.$$

a) Find bases for $\mathcal{N}(A)$ and $\mathcal{R}(A)$.

b) Find bases for $\mathcal{N}(A^T)$ and $\mathcal{R}(A^T)$.